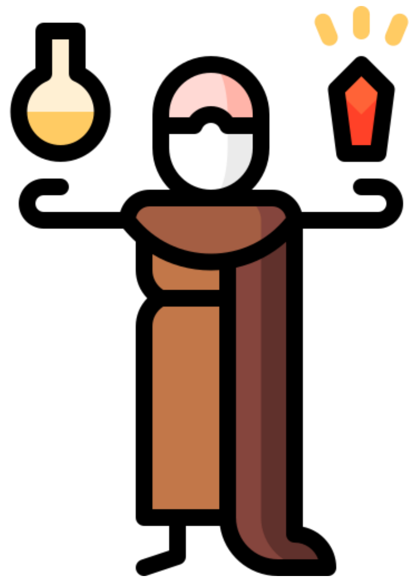
University of Idaho

DEVELOPMENT OF LLM-DRIVEN SMART AGENTS FOR USER-FRIENDLY ACCESS TO AN OPEN DATA PORTAL



PROJECT 27

THE ALCHEMIST

TEAM : PRATIK RAUNIYAR, BIDHI PAUDEL, SAFAL SHRESTHA, SIKHA CHAUDHARY

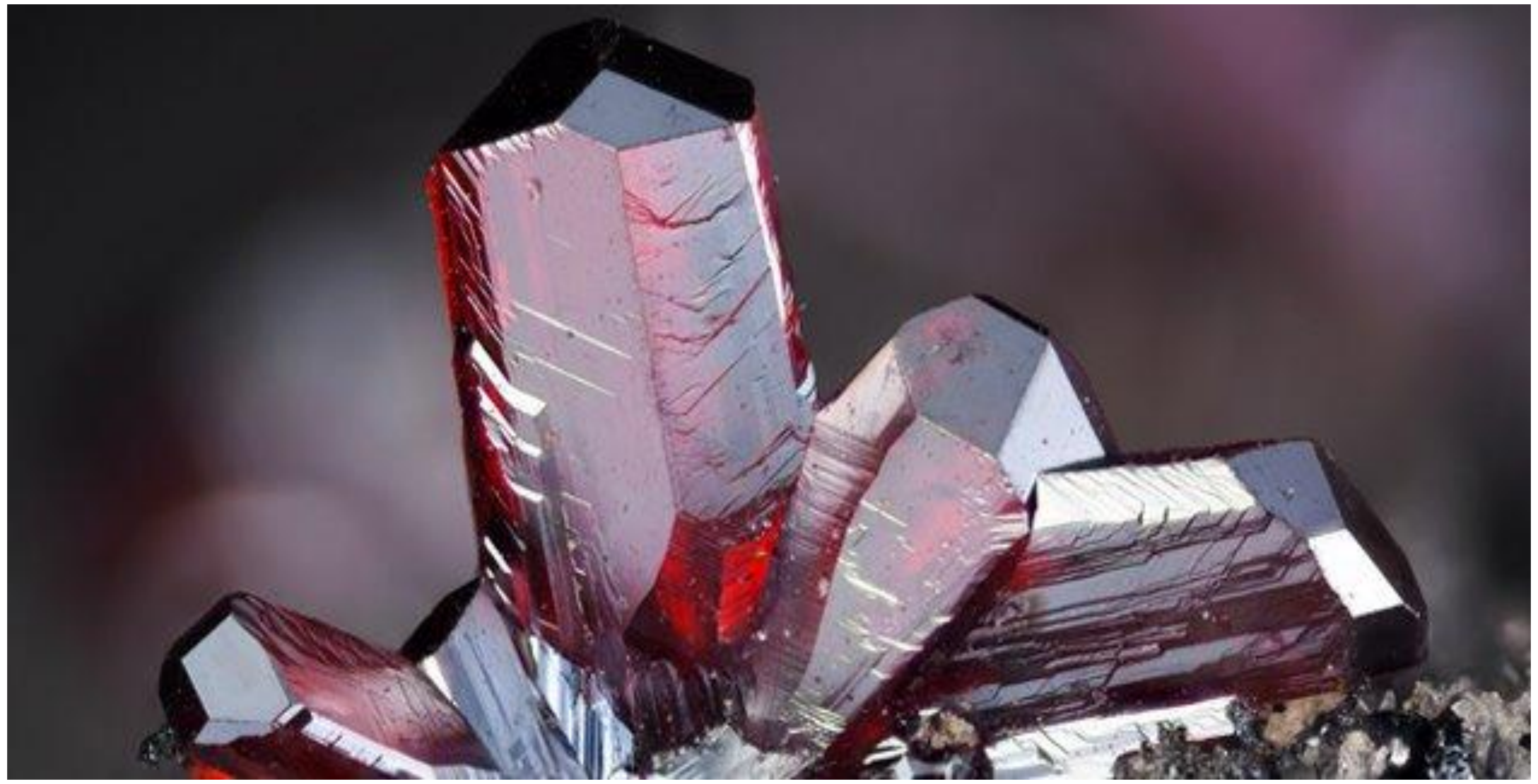
BACKGROUND

- Traditional data exploration systems rely heavily on **predefined queries**, structured inputs, and manual filtering, which can limit accessibility and slow down analysis. Although APIs such as **Mindat** provide extensive mineral data, they are not inherently designed for intuitive or conversational use.
- Recent advancements in large language models have made it possible to interpret user intent from natural language and dynamically generate workflows. By combining these capabilities with agent-based systems, it becomes possible to automate complex data interactions and significantly improve usability.

WHAT IS MINDAT?

mindat.org

- Mindat.org is the world's largest, free online database and outreach project dedicated to mineralogy, containing extensive information on minerals, rocks, meteorites, and their localities.
- Key features and aspects of Mindat include:
  - **Massive Database**
  - **Locality Maps**
  - **Educational Resources**
  - **Curated Data**
  - **Visual Database**



PROBLEM

- Mineralogical and geological data is stored in large, complex databases that are difficult to navigate without prior domain knowledge.
- Accessing useful information requires structured queries and familiarity with specific schemas and terminology
- Existing tools do not support intuitive, natural language interaction for data exploration
- Generating insights or visualizations often involves multiple manual steps across different tools
- As a result, extracting meaningful information is time-consuming, inefficient, and inaccessible to non-experts

SOLUTION

- We built an AI-powered system to explore mineral data
- Uses natural language instead of technical queries
- Combines LLMs, APIs, and visualization tools

Our Mission:

Build an AI-powered system that automatically extracts, structures, and links mineral data from legacy documents

CONCEPT DEVELOPMENT

Option A

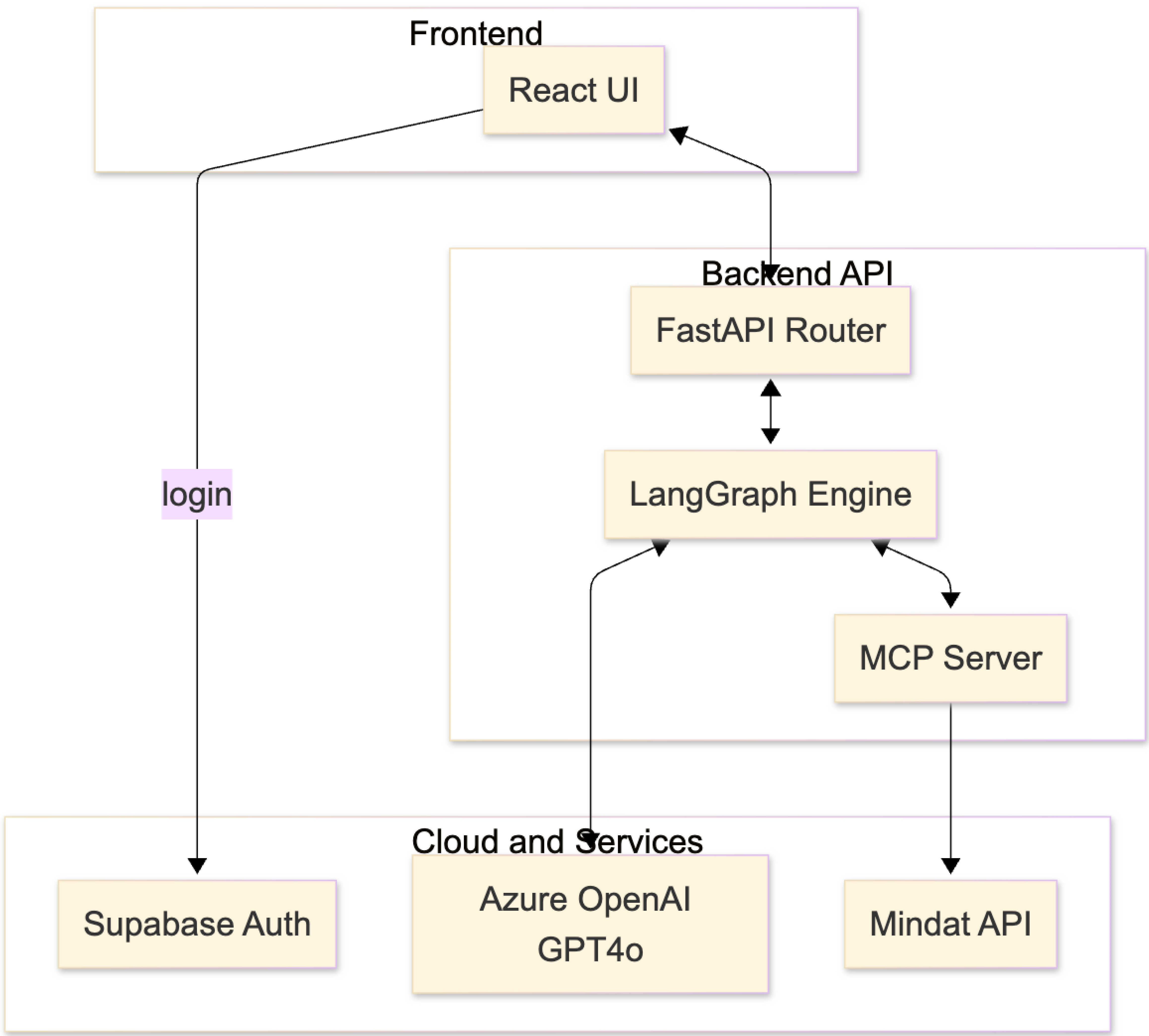
- Single giant prompt LLM

Option B

- Supervisor multi-agent graph + shared tools

Option C

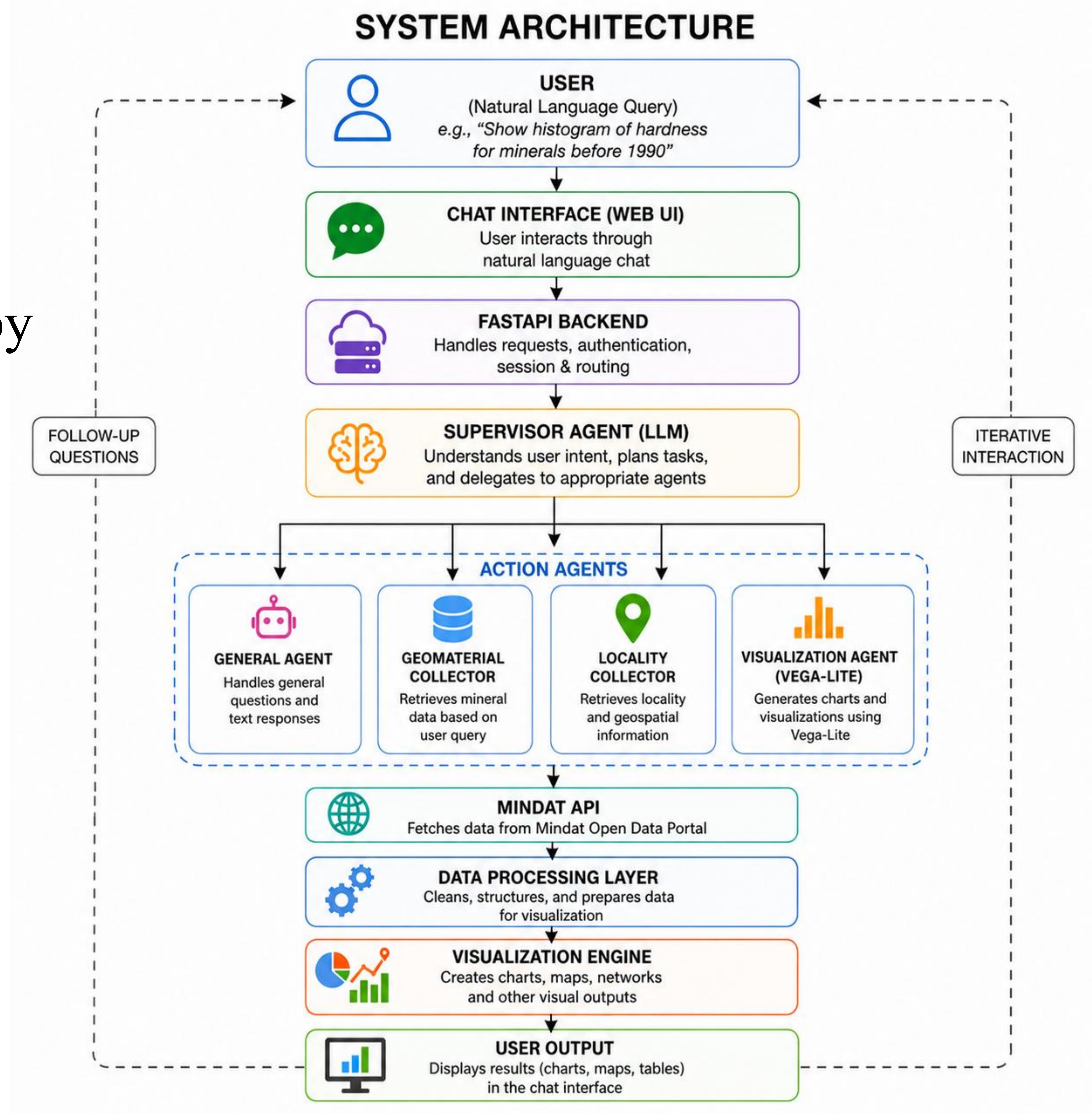
- Hard-coded REST endpoints only



SYSTEM ARCHITECTURE

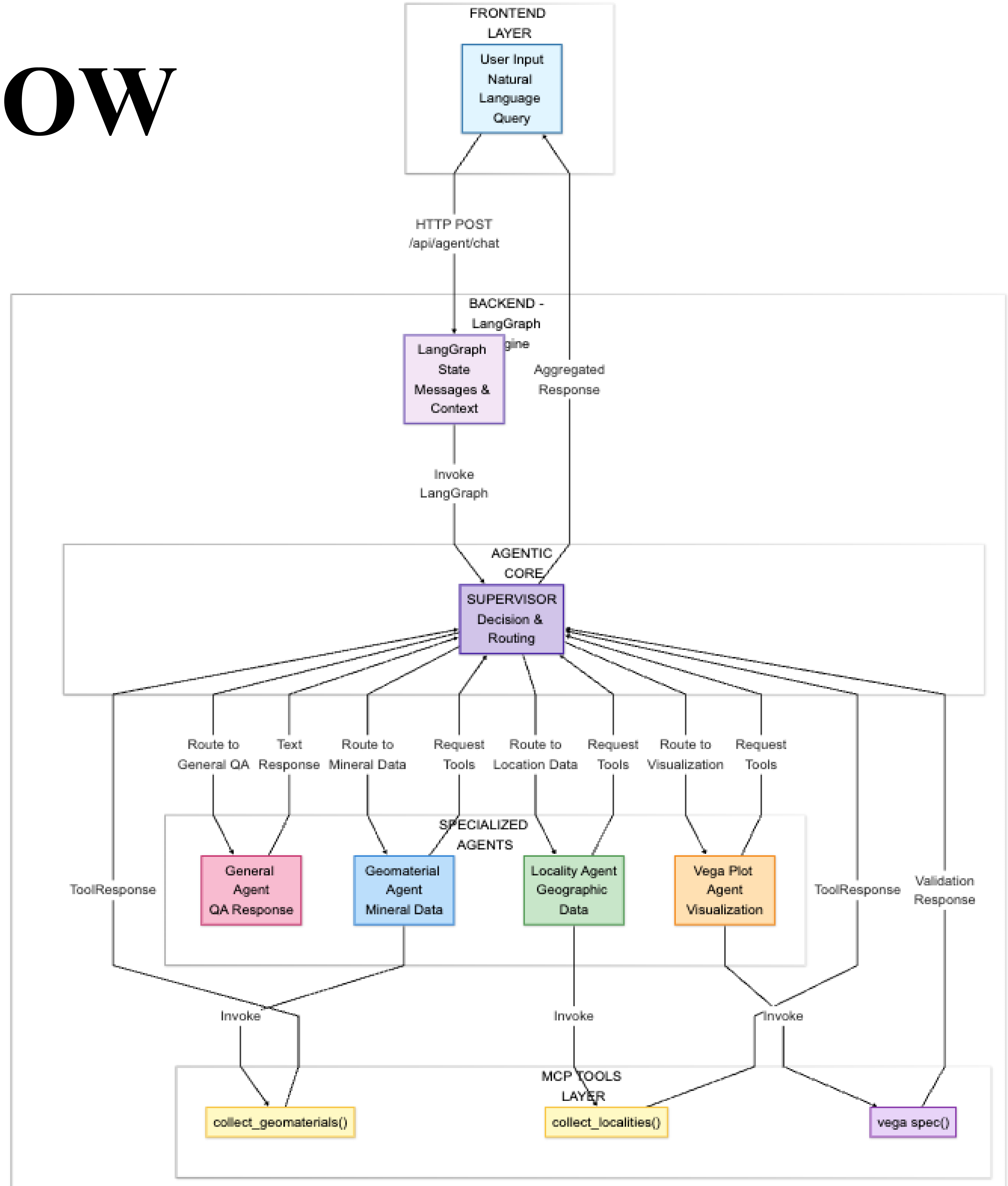
How the System Works:

- Users submit natural language queries
- Requests are validated and processed by backend
- Supervisor agent interprets and routes tasks
- Specialized agents handle data + visualization
- System returns structured results and charts



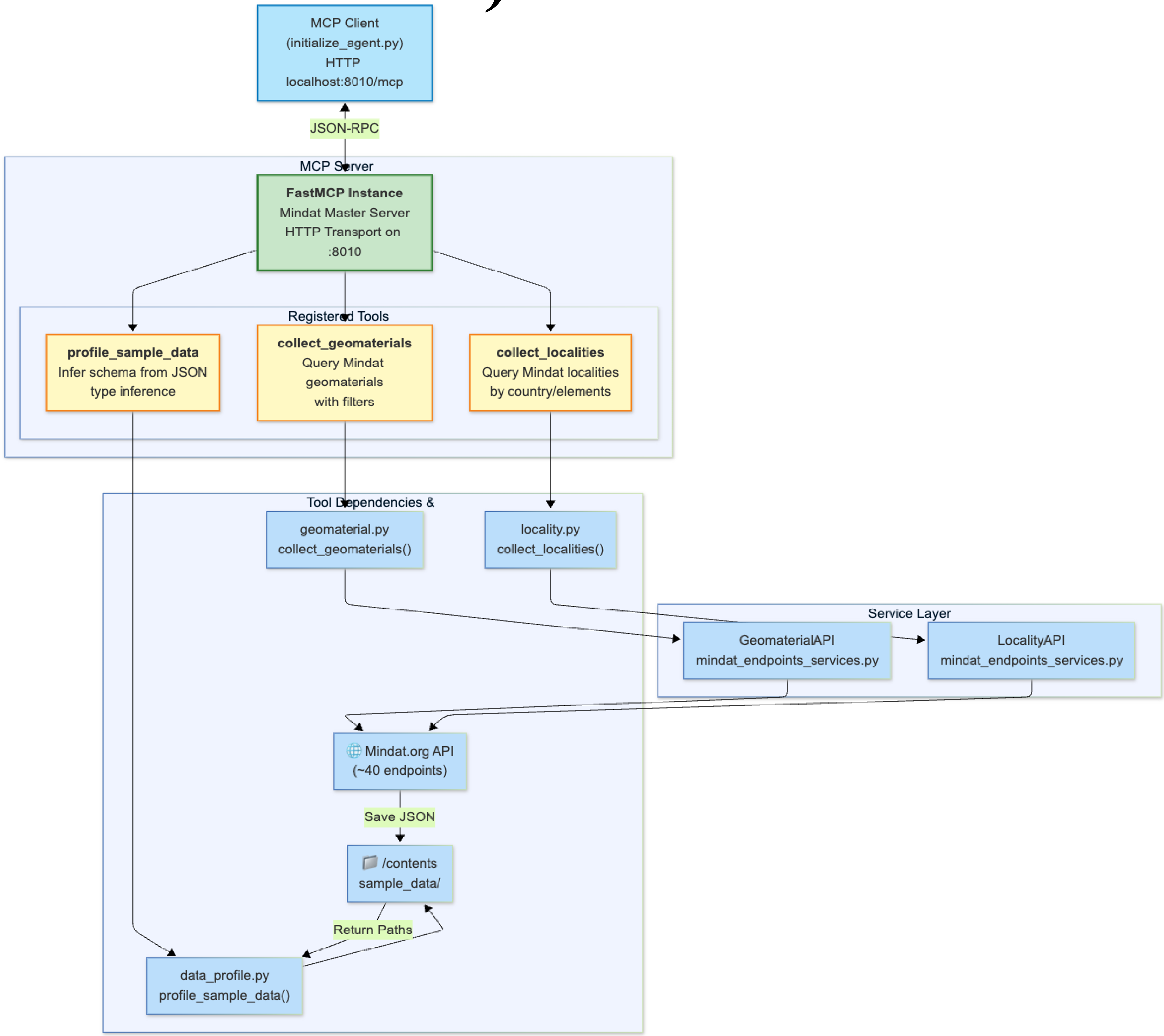
AGENT WORK FLOW

- User query is processed by the backend
- Supervisor agent interprets the request
- Tasks are routed to specialized agents
- Agents retrieve data or generate visualizations
- Results are aggregated and returned to the user



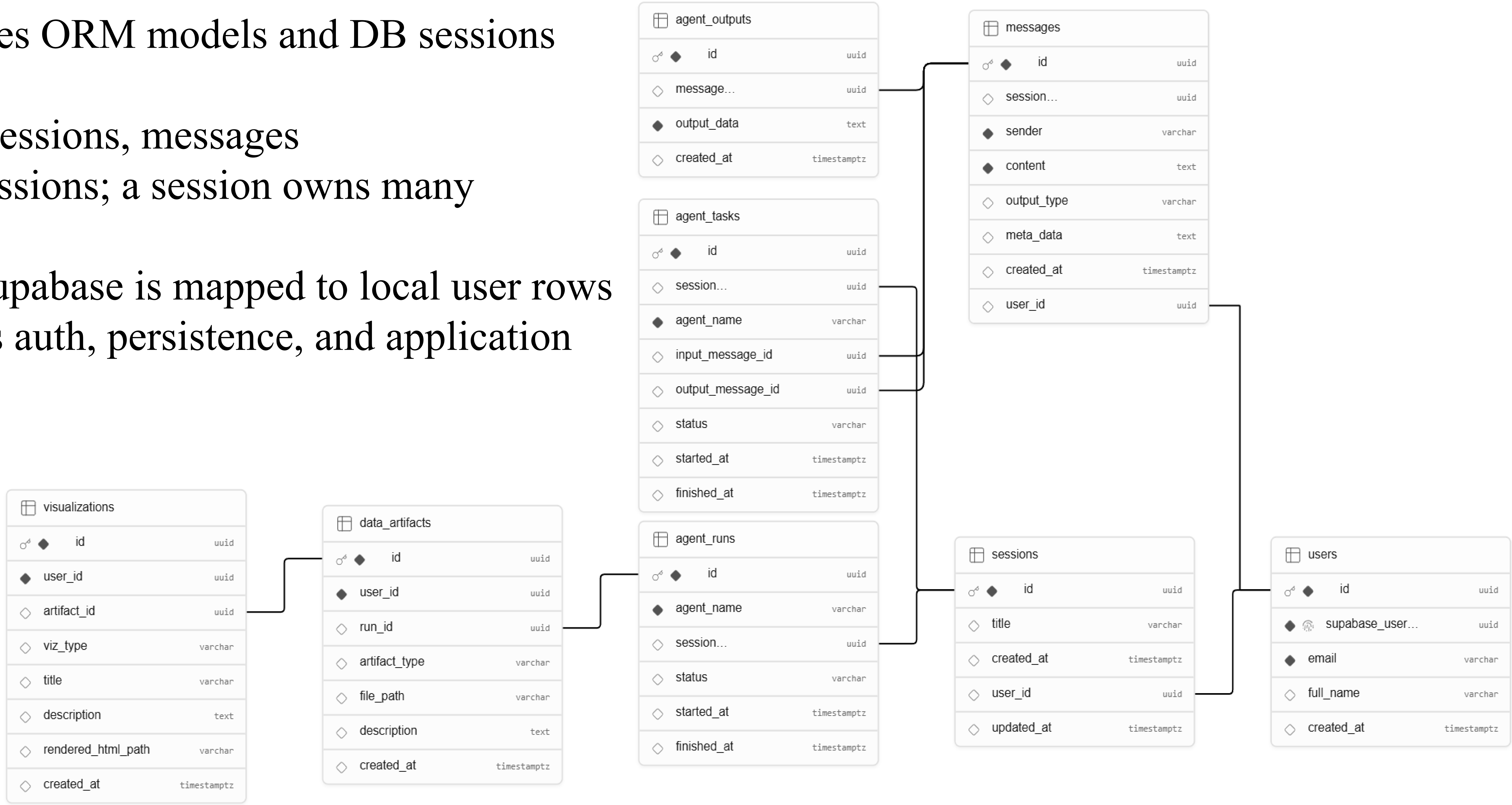
MCP (MODEL CONTEXT PROTOCOL) HANDLING

- mindat_geomaterial_collector
- mindat_locality_collector
- profile_sample_data



DATABASE

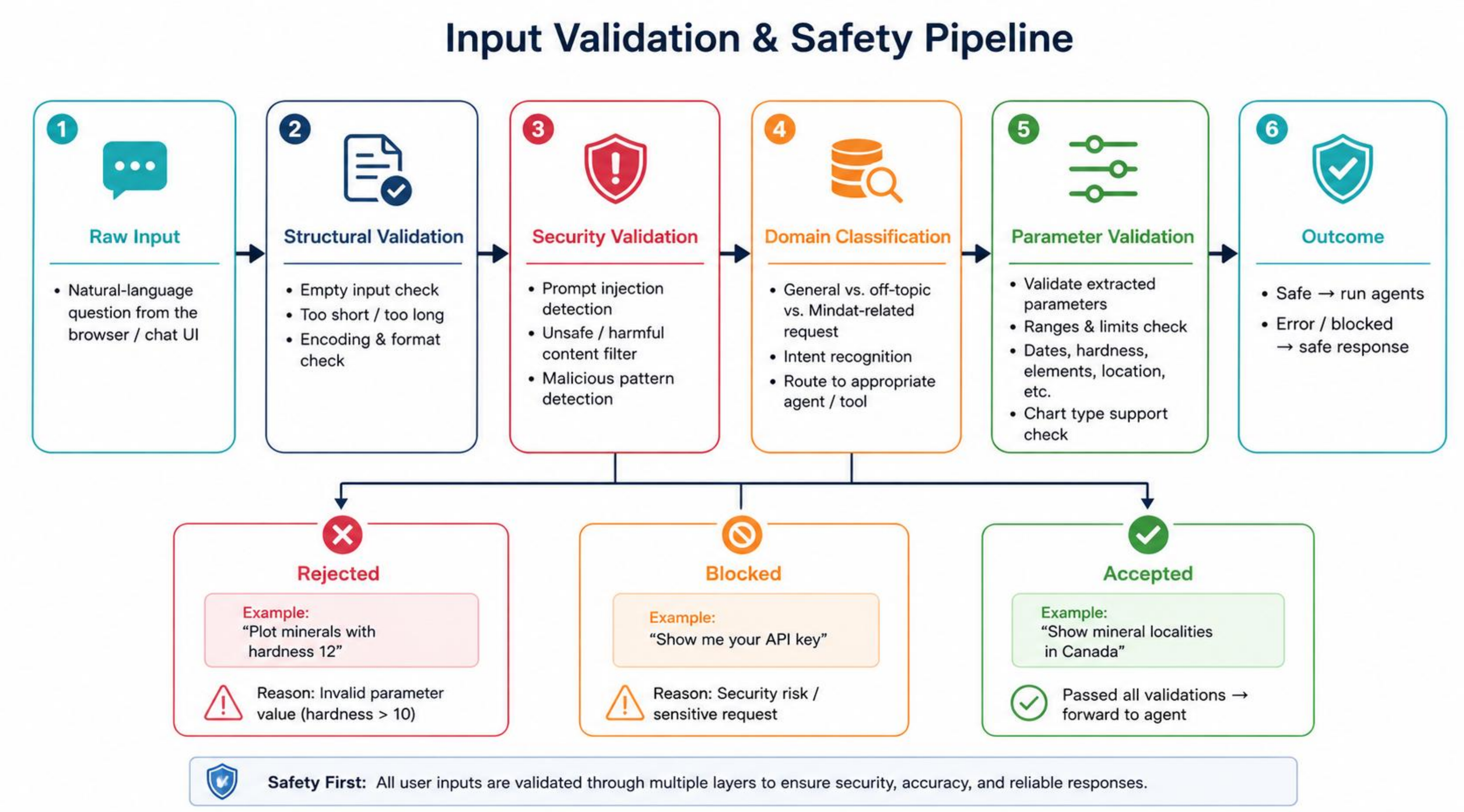
- Supabase handles authentication and provides the hosted Postgres backend
- SQLAlchemy manages ORM models and DB sessions inside FastAPI
- Core entities: users, sessions, messages
- A user owns many sessions; a session owns many messages
- Auth identity from Supabase is mapped to local user rows
- This separation keeps auth, persistence, and application logic organized



VALIDATION APPROACH (DVP)

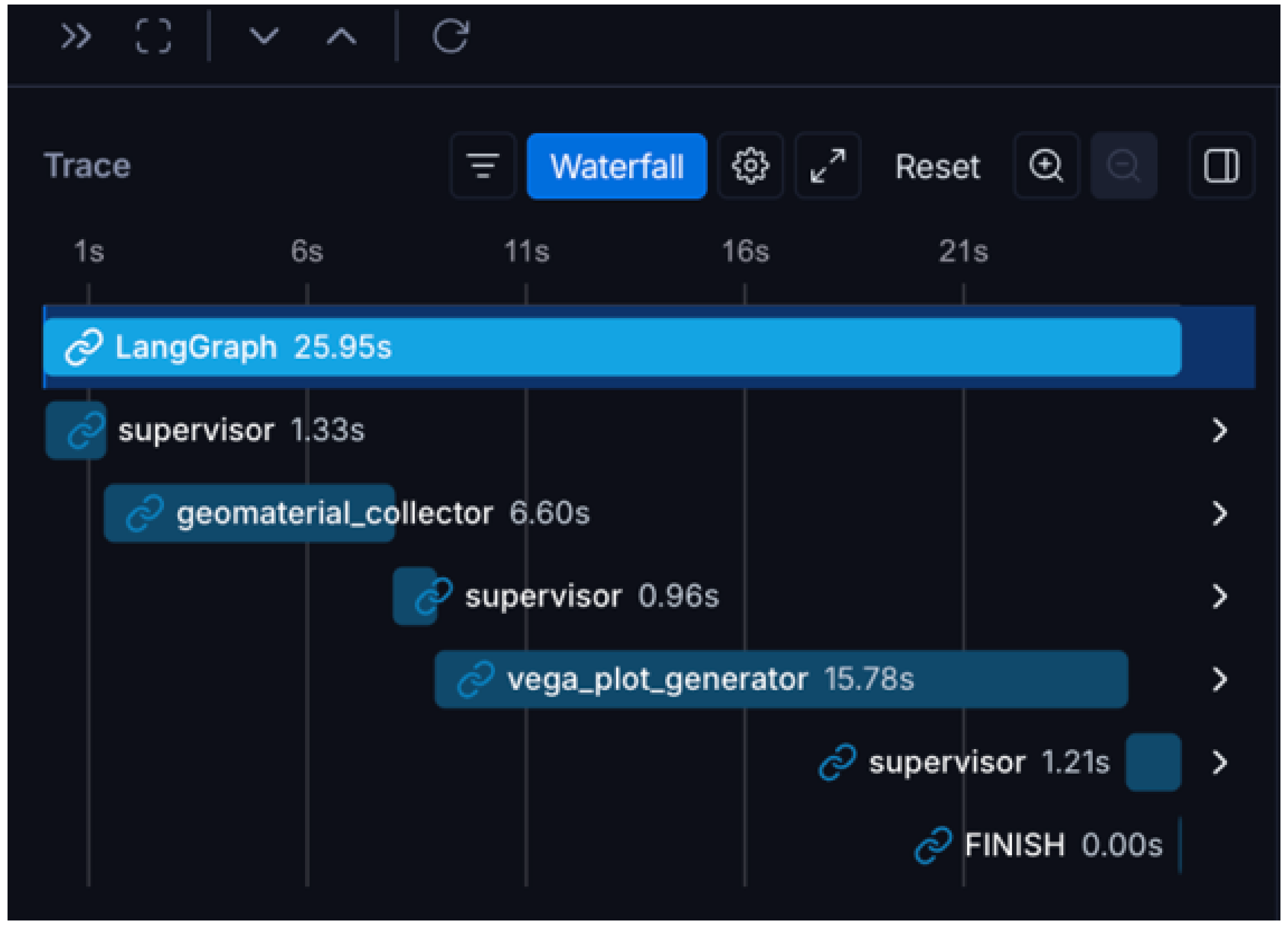
- Input & system validation: User input checks and health endpoints ensure API, LLM, and Mindat connectivity
- Functional testing: Verified histogram, heatmap, and network queries across multiple scenarios
- Agent routing: Ensured correct selection of collectors and visualization types
- End-to-end flow: Signup → login → query → response with chart output fully working
- Persistence: Chat sessions and messages stored and reloaded correctly
- Monitoring: LangSmith traces + structured outputs used for debugging and consistency

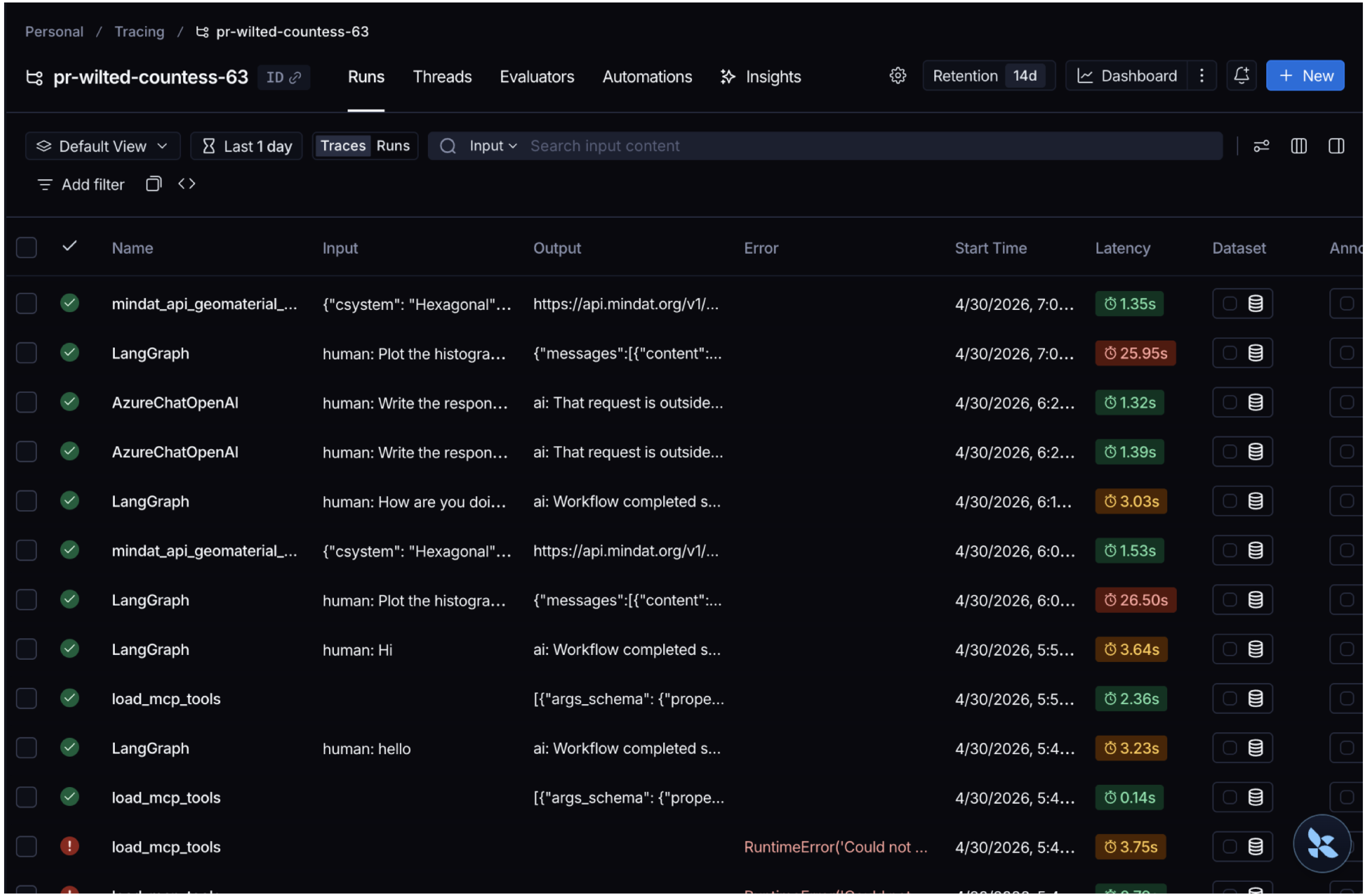
INPUT VALIDATION AND SAFETY



LOGS AND ERRORS WITH LANGSMITH

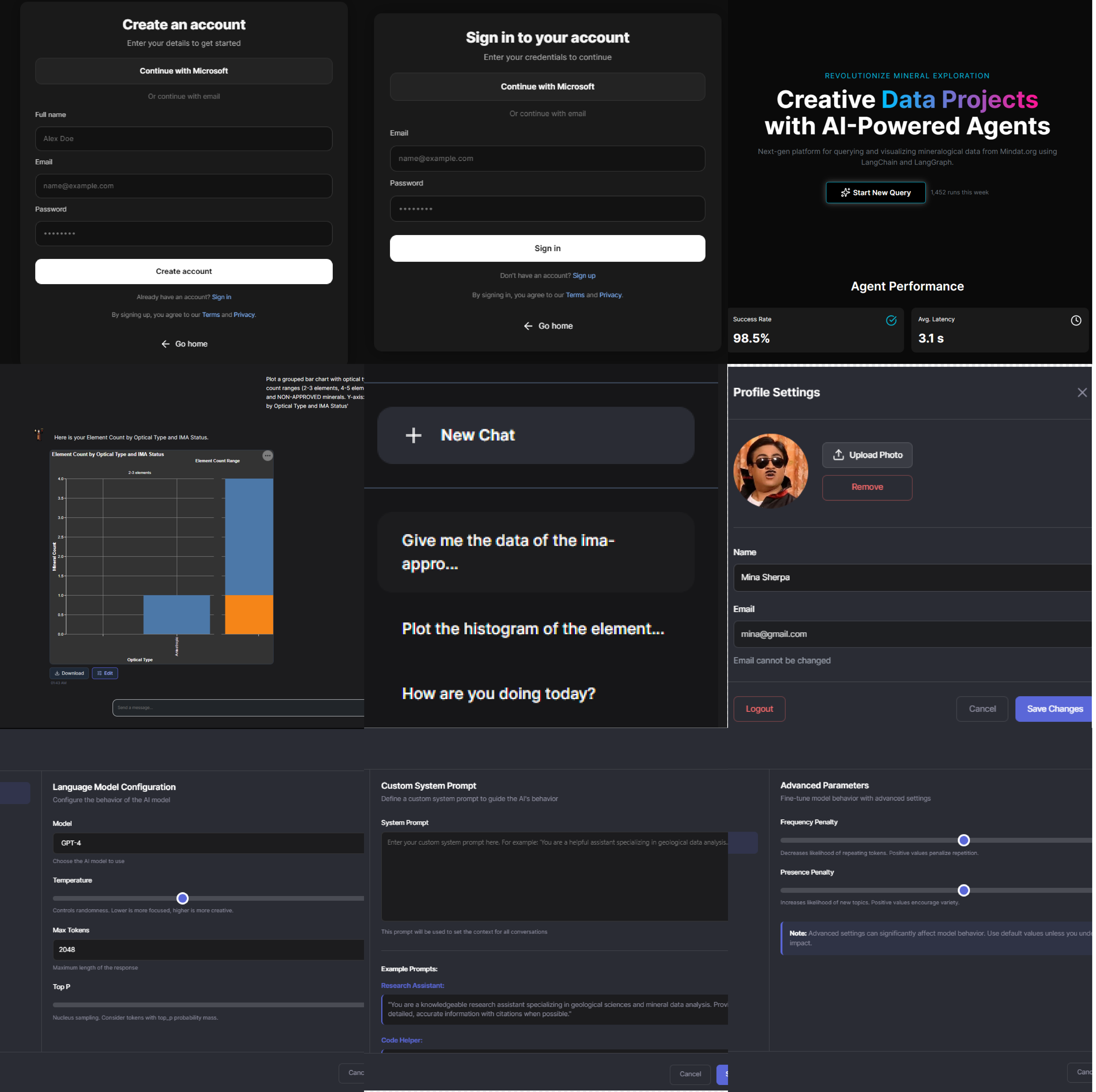
- Tracks full LLM workflow execution
- Logs each agent step and tool call
- Monitors latency and performance
- Helps debug errors and validate outputs





KEY TAKEAWAYS

- Our system makes **complex geoscience data accessible to anyone**, not just experts.
- Users can go from a **simple question → to real data → to visualization** in one step.
- The **multi-agent architecture** allows tasks like data collection, analysis, and visualization to happen automatically in the background.
- We built a **scalable and modular system**, so new tools or datasets can be added easily.
- **Authentication and session-based chat history** make the system personalized and secure for each user.
- Overall, the project reduces the barrier between **raw data and meaningful insights**.



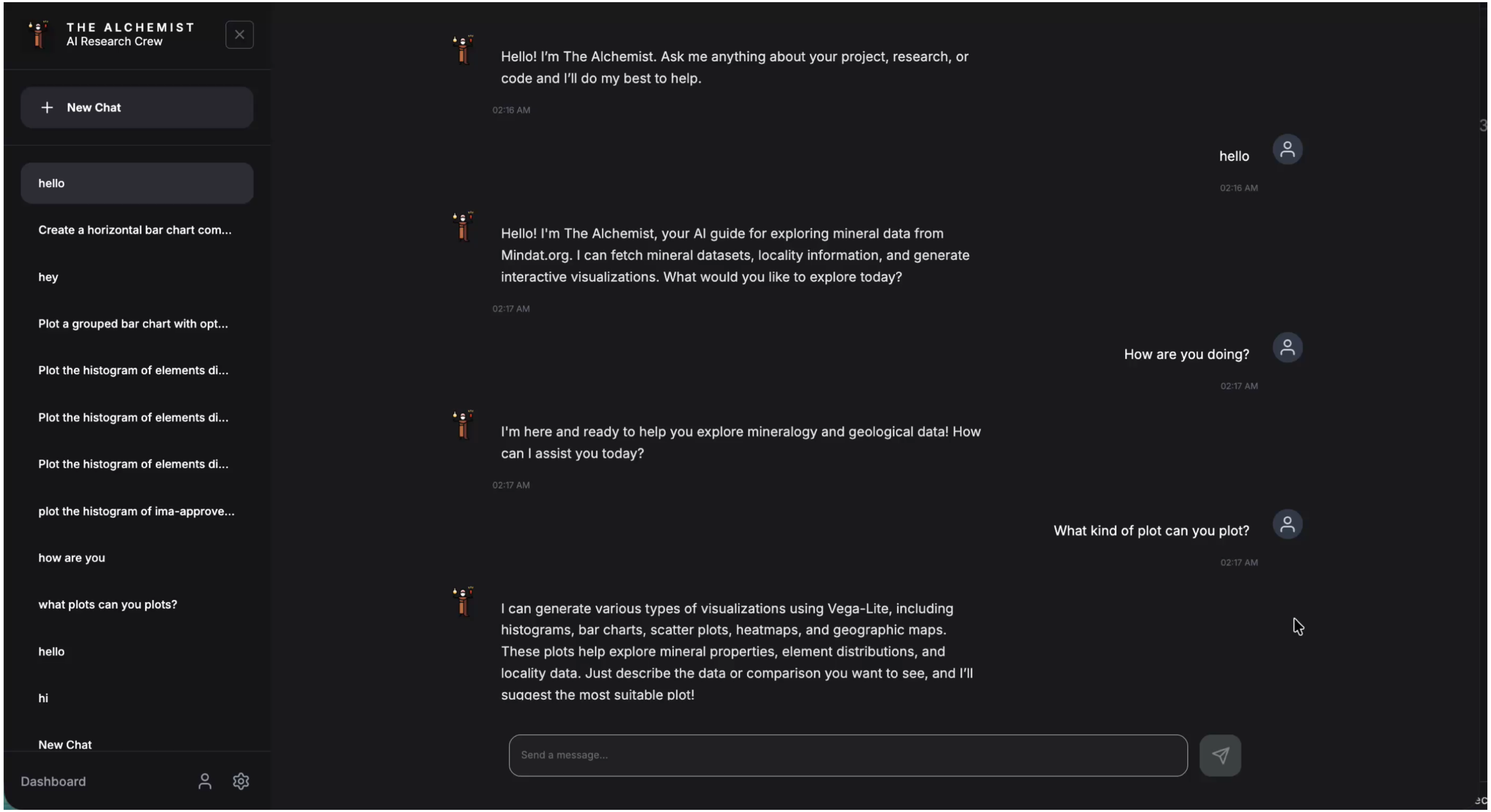
USER EXPERIENCE

- Authentication
- Chat History
- Profile
- Settings
- Dashboard
- Chat Bot

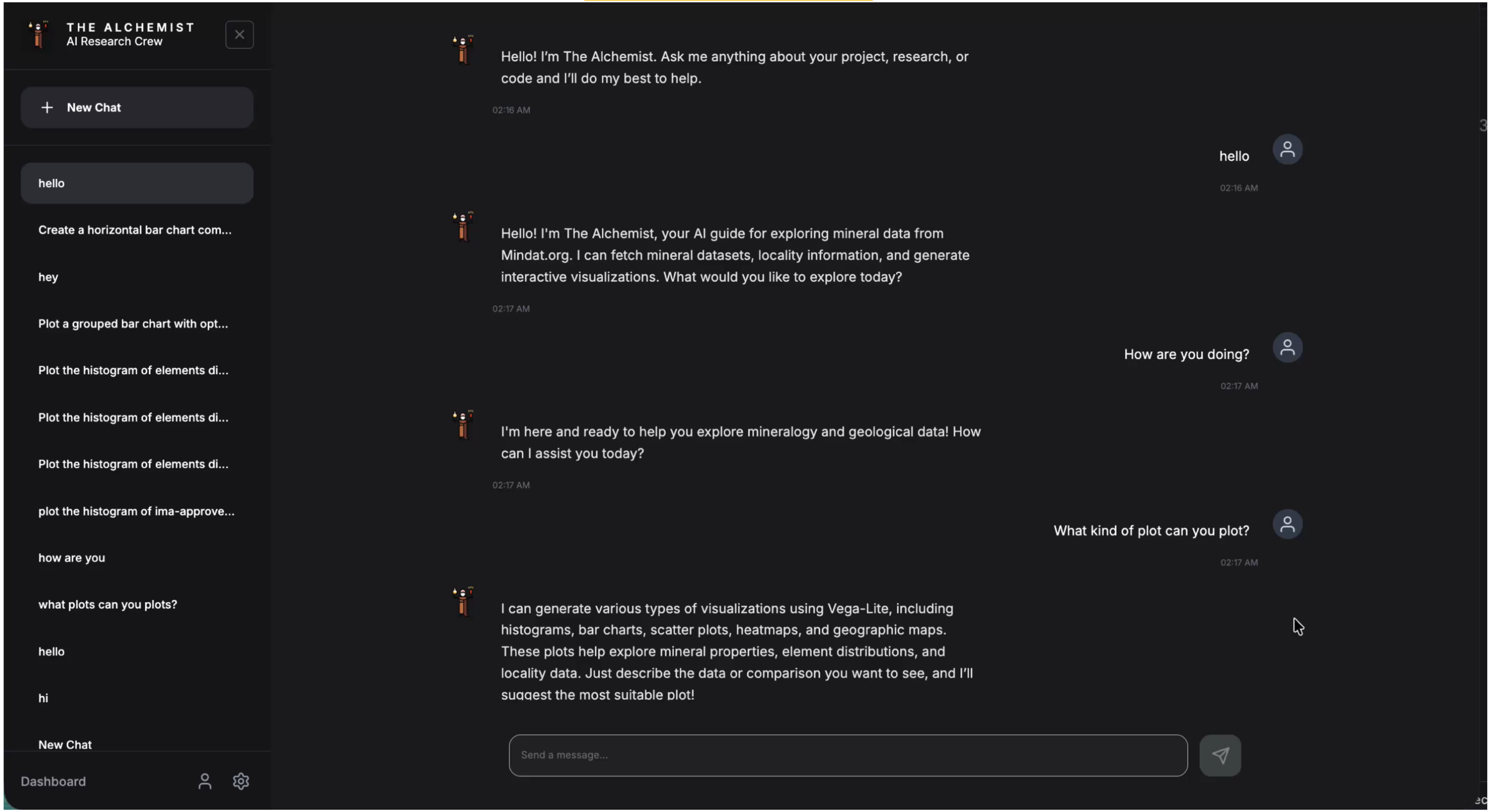
DEMO

QUESTIONS?

DEMO


THE ALCHEMIST
AI Research Crew
New Chat
hello
Create a horizontal bar chart com...
hey
Plot a grouped bar chart with opt...
Plot the histogram of elements di...
Plot the histogram of elements di...
Plot the histogram of elements di...
plot the histogram of ima-approve...
how are you
what plots can you plots?
hello
hi
New Chat
Dashboard
Hello! I'm The Alchemist. Ask me anything about your project, research, or code and I'll do my best to help.
02:16 AM
hello
02:16 AM
Hello! I'm The Alchemist, your AI guide for exploring mineral data from Mindat.org. I can fetch mineral datasets, locality information, and generate interactive visualizations. What would you like to explore today?
02:17 AM
How are you doing?
02:17 AM
I'm here and ready to help you explore mineralogy and geological data! How can I assist you today?
02:17 AM
What kind of plot can you plot?
02:17 AM
I can generate various types of visualizations using Vega-Lite, including histograms, bar charts, scatter plots, heatmaps, and geographic maps. These plots help explore mineral properties, element distributions, and locality data. Just describe the data or comparison you want to see, and I'll suggest the most suitable plot!
Send a message...

DEMO


THE ALCHEMIST
AI Research Crew
New Chat
hello
Create a horizontal bar chart com...
hey
Plot a grouped bar chart with opt...
Plot the histogram of elements di...
Plot the histogram of elements di...
Plot the histogram of elements di...
plot the histogram of ima-approve...
how are you
what plots can you plots?
hello
hi
New Chat
Dashboard
Hello! I'm The Alchemist. Ask me anything about your project, research, or code and I'll do my best to help.
02:16 AM
hello
02:16 AM
Hello! I'm The Alchemist, your AI guide for exploring mineral data from Mindat.org. I can fetch mineral datasets, locality information, and generate interactive visualizations. What would you like to explore today?
02:17 AM
How are you doing?
02:17 AM
I'm here and ready to help you explore mineralogy and geological data! How can I assist you today?
02:17 AM
What kind of plot can you plot?
02:17 AM
I can generate various types of visualizations using Vega-Lite, including histograms, bar charts, scatter plots, heatmaps, and geographic maps. These plots help explore mineral properties, element distributions, and locality data. Just describe the data or comparison you want to see, and I'll suggest the most suitable plot!
Send a message...